MIL-E-36478B
20 October 1976
SUPERSEDING
MIL-E-36478A
27 September 1971

MILITARY SPECIFICATION

ENVELOPE, STERILIZATION, PAPER, SELF-SEALING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers disposable paper envelopes for use in holding hypodermic needles and dental syringes during autoclaving procedures at 250° F. and 15 psi, and to maintain sterility thereafter.

1.2 Classification.

* 1.2.1 Types. The envelopes shall be of the following types, as specified (see 6.2):

Type I Envelope, sterilization, paper, self-sealing, 3-5/8 by 1-5/8 inches.

Type II Envelope, sterilization, paper, self-sealing, 8 by 2-3/4 inches.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6530

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI STANDARDS

T-400 - Sampling a Single Lot of Paper, Paperboard, Fiberboard or Related Product.
T-403 - Bursting Strength of Paper.
T-404 - Tensile Breaking Strength of Paper and Paperboard.
T-410 - Basis Weight of Paper and Paperboard.
T-412 - Moisture in Paper and Paperboard.
T-414 - Internal Tearing Resistance of Paper.
T-425 - Opacity of Paper

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 1 Dunwoody Park, Atlanta, GA 30341.)

U. S. PHARMACOPEIAL CONVENTION, INC. (USP)

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.

3.1.1 Paper. The paper shall be made from 100 percent fully bleached chemical wood pulp, white color, having the characteristics listed in table I, when tested as specified in 4.4.5.

TABLE I. Characteristics of paper.

Characteristic	Unit	Required value
Basis weight	pounds	30 minimum
Moisture content	percent	3 to 7
Tear strength	grams	
Machine direction		40 minimum
Cross direction		40 minimum
Mullen strength (dry)	pounds per square inch	20 minimum
Tensile strength (dry)	pounds per inch of width	
Machine direction		20 minimum
Cross direction		10 minimum
Opacity	percent	70 minimum

The paper shall be sufficiently permeable to permit the entrance of live steam during sterilization, and sufficiently impermeable after sterilization so that contamination from the atmosphere shall not enter the closed envelope.

3.1.2 Adhesives.

* 3.1.2.1 Open-end flap. The adhesive used for closure of the open-end flap shall be of the self-sealing type of latex gum and shall be capable of adhering only to an adhesive of similar composition. The adhesive material shall be capable of withstanding sterilization and maintaining sterility as specified. The adhesive shall be capable of providing a proper seal and shall show no deterioration or increased tackiness when tested as specified in 4.4.2. Adhesive shall have a shelf life of not less than 36 months.

3.1.2.2 Center and bottom seams. The adhesive material used for sealing the center and bottom seams shall be a good quality adhesive, capable of withstanding sterilization and maintaining sterility as specified in 3.4 and 3.5.

* 3.2 Construction. The style of the envelopes shall consist of a center and bottom seam with a safety fold bottom. Both seams shall be completely sealed. The minimum overall dimensions of the envelopes shall be as specified in table II.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:40Z
Check the source to verify that this is the current version before use.

TABLE II. Minimum dimensions of envelopes (inches)

Dimensions	Type I	Type II
Length (with open-end flap closed and closed-end flap closed)	3-1/2	7-7/8
Width	1-1/2	2-5/8
Width of center seam overlap	1/2	5/8

* 3.2.1 Flaps. The open-end flap and the closed-end flap (which forms the bottom seam), shall be designed to provide a safety fold of not less than 3/16 inch across the entire width of the envelope when they are folded over the body of the envelope. This shall be facilitated by inclusion of a crease located a minimum of 3/16 inch from the top and bottom edges of the back of the envelope. The open-end flap shall measure at least 1-1/8 inches including the safety fold. The bottom sealed flap shall measure at least 15/16 inch including the safety fold. The finished envelope shall be supplied with the open-end flap in the open position and the bottom flap closed to form the bottom seam.

* 3.2.2 Self-sealing adhesive. The self-sealing adhesive material specified in 3.1.2.1 shall be applied to the open-end flap and the body of the envelope in strips measuring a minimum of 1-1/8 inches long by 1/2 inch wide for type I envelopes and 2 inches long by 1/2 inch wide for type II envelopes. The adhesive strips, on the envelope and open-end flap, shall be so positioned that when the flap is closed (folded along the crease as specified above) the two adhesive strips shall coincide and effect a permanent seal, by means of finger pressure, for the contents of the envelope. A spot or dab of adhesive material at these points shall not be acceptable.

3.3 Marking.

* 3.3.1 Indicator marking. Each envelope shall be marked with a time-temperature-steam-sensitive ink which will change color (readily discernible color change required) under conditions for sterilizing, providing a positive indication that the contents have been subjected to conditions of sterilization. When tested as specified in 4.4.1, there shall be a visible color change in the autoclave indicator ink. When tested as specified in 4.4.4, there shall be no visible color change in the autoclave indicator ink.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

3.3.2 Printing. The following printing shall be placed on the type II envelope only.

Syringe _____

Needle(s) _____

Other _____

_____

Date _____

Initials _____

The horizontal lines shall be placed at least 1/2 inch apart. The words "open this end" shall be printed on the upper flap of the envelope.

* 3.4 Sterilization. The closed envelope with contents enclosed shall be capable of withstanding two successive sterilization periods as specified in 4.4.1, without deterioration of the envelope or of the adhesive, and without opening of the envelope. In addition, for type II envelopes, all printing shall meet with the requirement of 3.3.1 (readily discernible color change) during the first sterilization cycle. As an alternate, a one inch by one inch minimum size symbol shall turn color in lieu of the printing.

* 3.5 Maintenance of sterility. When tested as specified in 4.4.3, the contents of the envelopes tested immediately after sterilization, and those tested 30 days later, shall show no evidence of bacterial or mold growth.

3.6 Workmanship. Workmanship shall be first class throughout. Envelopes shall be free from defects which detract from their appearance or may impair their serviceability. They shall contain no cuts, tears, or holes. They shall show no evidence of grease, oil, or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table II. The unit of product for examination shall be one envelope. Not more than 25 envelopes shall be taken at random from each unit of issue.

TABLE III. Sampling for examination.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

* 4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table IV. Acceptance number shall be zero for all sample sizes.

TABLE IV. Sampling for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level
Paper	Physical Characteristics	3.1.1	4.4.5	1/
Envelope	Adhesive durability	3.1.2.1	4.4.2	S-1
	Dry heat	3.3.1	4.4.4	S-1
	Sterilization	3.4	4.4.1	S-1
	Maintenance of sterility	3.5	4.4.3	2/

1/ Samples of paper, used in the fabrication of envelopes, shall be taken in accordance with TAPPI method 400 and the TAPPI test methods listed in table VI.

2/ Twenty (20) envelopes shall be selected from each lot for sterility testing for first article only.

4.3 Examination. The envelopes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with classification of defects shown in table V. Examination shall not be restricted to the classified possible defects listed below.

TABLE V. Classification of defects.

Categories	Defects
Major	
101	Envelope not free of tears, cuts and holes.
102	Envelope not free of grease, oil, and foreign matter.
103	Center or bottom seam, or both, not completely sealed.
104	Adhesive strip(s) missing.
105	Adhesive strips do not coincide for sealing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

TABLE V. Classification of defects - Continued

Categories	Defects
Minor	
201	Envelope not free of ripples and unspecified creases.
202	Printing (type II only) missing, incomplete or illegible.
203	Excessive adhesive used in sealing the center or bottom seam or both.
204	Crease in flap missing.

4.3.2 Dimensions. The envelopes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Sterilization test. Sterilization shall be at 121° Celsius (C) and a steam pressure of 15 pounds per square inch for 30 minutes. Sterilization shall be repeated in accordance with the requirements of 3.4.

4.4.2 Adhesive oven test. Properly sealed envelopes shall be placed in a circulating air oven at 75° C for 15 days. At the end of this time, the envelopes shall be examined for compliance with 3.1.2.1.

* 4.4.3 Sterility test. Twenty type I envelopes, each containing a spore strip impregnated with Bacillus Stearothermophilus, and a hypodermic needle (or twenty type II envelopes, each containing a similarly impregnated spore strip and a dental syringe) shall be properly sealed and sterilized once, as specified in 4.1.1. Immediately after sterilization, the contents of ten envelopes shall be tested as specified below. The remaining ten envelopes shall be exposed to the atmosphere at ambient temperature (18° C to 27° C) for 30 days and the contents shall then be tested as specified below. To test for sterility, aseptically transfer five of the needles and all ten of the spore strips to culture tubes containing fluid soybean casein digest. Incubate the needles at 20° C to 25° C and incubate the spore strips at 55° C to 65° C for not less than seven days. Transfer the five remaining needles to culture tubes containing fluid thioglycollate medium and incubate at 30° C to 35° C for not less than seven days. The quantity and preparation of the media shall be as described under sterility tests in the U.S.P.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

* 4.4.4 Dry heat test. Envelopes shall be placed in a dry heat oven for 30 minutes at 104° C + 3° C. The envelopes shall be removed and allowed to cool for 30 minutes and then examined for color change.

* 4.4.5 Physical characteristics of paper. Paper shall be tested in accordance with applicable method shown in table VI and referenced in TAPPI testing procedures for compliance with 3.1.1.

TABLE VI. Tests for physical characteristics of paper.

Characteristic	TAPPI test procedure
Basis weight	T 410
Moisture content	T 412
Tear strength	T 414
Bursting (Mullen) strength	T 403
Tensile strength	T 404
Opacity	T 425

4.5 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

* 5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One box containing five hundred type I or type II envelopes, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Five hundred envelopes of one type only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Five hundred envelopes of one type only shall be clean and packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:40Z
Check the source to verify that this is the current version before use.

5.2.1 Level A. Thirty-six unit packages containing type I envelopes or ten unit packages containing type II envelopes shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Thirty-six unit packages containing type I envelopes or ten unit packages containing type II envelopes shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged envelopes shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Unit package and exterior container. Each unit package and exterior container shall be marked in accordance with MIL-STD-129. Type II shelf life markings as specified in MIL-STD-129 shall be shown.

6. NOTES

* 6.1 Intended use. Envelopes covered by this specification are intended to assure and maintain sterility of hypodermic needles (type I envelopes) and dental syringes (type II envelopes).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Type required.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:40Z
Check the source to verify that this is the current version before use.

6.3 This specification covers the following items listed in the Federal Supply Catalog.

National Stock Number	Item Identification	Type
6530-00-299-8370	ENVELOPE, STERILIZATION, PAPER, SELF-SEALING, 500s. 3-5/8 by 1-5/8 inches, for hypodermic needles.	I
6530-00-754-0421	ENVELOPE, STERILIZATION, PAPER, SELF-SEALING, 500s, 8 by 2-3/4 inches for dental syringes.	II

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6530-1351

*U.S. GOVERNMENT PRINTING OFFICE: 1974 703-020 1-41

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:40Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R-254

INSTRUCTIONS - The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers, vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-E-36478B ENVELOPE, STERILIZATION, PAPER, SELF-SEALING

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.
   B. RECOMMENDATION FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:40Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:40Z
Check the source to verify that this is the current version before use.